25900201 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCSAC100 - MUSCULOSKELETAL SYSTEM AND BIOMECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What is the primary function of the muscles in the trunk? [U] / [CO1]

   a). Movement of the head | b). Movement of the arms and legs | c). Protection and support for internal organs | d). Regulating body temperature

2. Which of the following nerves innervates the biceps brachii muscle? [An] / [CO3]

   a). Femoral nerve b). Median nerve

   c). Musculocutaneous nerve d). Radial nerve

3. Which plane of the body is most commonly used in imaging studies like MRIs and CT scans? [C] / [CO5]

   a). Transverse plane b). Coronal plane

   c). Sagittal plane d). All planes are used equally

4. At what muscle length is the maximum force typically generated according to the length-tension relationship? [U] / [CO6]

a). At the muscle's shortest length b). At the muscle's longest length

c). At an optimal length where actin and myosin overlap maximally d). At an intermediate length with no overlap of actin and myosin

5. What is a motor unit? [U] / [CO6]

a). A single muscle fiber activated by multiple motor neurons. b). A group of muscle fibers activated by a single motor neuron.

c). A type of muscle fiber involved in fast contractions. d). A group of muscle fibers responsible for sustained contractions.

6. Which part of the skeletal muscle is responsible for contraction? [K] / [CO6]

a). Sarcoplasmic reticulum b). Myofibrils

c). Nucleus d). Mitochondria

7. The coronal plane divides the body into which two parts? [C] / [CO5]

a). Upper and lower parts b). Left and right halves

c). Front and back halves d). Internal and external halves

8. Which of the following principles explains the relationship between the force applied to an object and its resulting motion? [U] / [CO4]

a). Newton's third law of motion b). Work-energy principle

c). Newton's second law of motion d). Principle of conservation of angular momentum

9. What function of the skeletal system is observed when an anatomical model shows the rib cage surrounding the heart and lungs? [An] / [CO3]

a). Movement b). Protection of internal organs c). Storage of minerals d). Blood cell production

10. What is the main function of smooth muscle tissue? [U] / [CO1]

a). Movement of limbs b). Contraction of the heart c). Regulation of internal organ movements d). Protection of internal organs

[1x10 = 10]

Part B

Short Answer

Answer any **10** questions

Each question carries **1** marks

11. Assess why the gluteus maximus is essential for sports performance. [U] / [CO1]
12. Describe the function of the sternocleidomastoid muscle. [U] / [CO1]
13. Differentiate between isometric and isotonic contractions in terms of movement. [K] / [CO2]
14. How do bones help in movement? [U] / [CO2]
15. What are joints, and what is their function? [K] / [CO3]
16. Describe the length-tension relationship in muscles and relate it to optimal force production. [U] / [CO1, CO2]
17. Evaluate the role biomechanics plays in enhancing athletic performance. [An] / [CO4]
18. Generate a biomechanics-based strength training program for improving muscular power. [An] / [CO4]
19. Explain how muscle force contributes to movement in human motion. [C] / [CO5]
20. Examine the role of Type IIb muscle fibers in explosive movements and power-based sports. [U] / [CO6]
21. Describe the function of the myelin sheath and its role in neural transmission. [U] / [CO5]
22. Assess the effect of joint angle on muscle force production and provide practical applications in strength training. [U] / [CO6]

[1x10 = 10]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

23. Compare the role of voluntary and involuntary muscles in maintaining body functions and supporting athletic performance. [U] / [CO1]
24. Discuss the significance of isokinetic contractions in rehabilitation and athletic performance. [K] / [CO2]
25. Define joints and explain the different types of joints in the human body with examples. [U] / [CO2]
26. Compare the force-length and force-velocity relationships and their impact on muscle performance in training. [U] / [CO4]

27. Evaluate why understanding neural control of muscles is essential for optimizing sports performance. [C] / [CO5]

28. Examine how joint angle influences strength output in different exercises. [K] / [CO6]

[5x4 = 20]